

Zaki Fathi Fahrizal

Sydney, NSW | zakifathi987@gmail.com | +61 430 095 891 | LinkedIn: Zaki Fahrizal

ABOUT

3rd Year Mechanical Engineering student at UNSW with experience in CFD, FEA, CAD design, and thermal-fluid systems. Interested in Building (HVAC), Process, Energy systems, and Simulation-driven Engineering roles.

EDUCATION

UNIVERSITY OF NEW SOUTH WALES, Sydney **Sep 2023 – Present**
B. Engineering (Honours), Mechanical Engineering GPA: 3.64/4.00 | WAM: 76.67/100
Indonesia Maju Scholarship Awardee (Fully Funded Study) by Indonesia Ministry of Education and Culture

WORK/TEAM EXPERIENCE

Mechanical Engineering Intern (Production Plan) | *PT Semen Baturaja Tbk* **Dec 2025 – Mar 2026**
Thermal systems, CFD, FEM, Process Eng

- Determine daily Ingredients for Cement Production based on the Kiln Daily Performance to produce 5000 ton/day clinker production target.
- Daily Inspection on Refrigeration Performance, Kiln Temperature, and Air Quality from Control-Room and Onsite inspection for efficient production.
- Perform Mapping and Analysis on Rotary Kiln Suspension Pre-Heater to Minimize Carbon Production.

Simulation Team | *UNSW Bionics Heart* **Feb 2026 – Present**
CFD, CAD, Ms. Excel

- Perform CFD simulations to analyse blood-flow dynamics and support artificial heart design optimization.
- Develop flow-domain geometries in CAD and apply simulation assumptions/boundary conditions to reduce hemolysis risk. My main Role is to simulate the TAH using UDF to simulate the TAH operating on 5 L/min blood flow-rate.

Mechanical Engineer | *sUNSWim – UNSW ROV Team* **Aug 2025 – Present**
CAD, FEA, CFD, prototyping

- Design 12 ROV chassis and peripheral components for underwater operation and 1 m depth prototype testing.
- Performed 3-D Printing/Laser Cut manufacture test, CFD and structural FEA on design to ensure its safety during operation.

Director of Media and Communication | *PPIA (Indonesian Student Association of Australia)* **Jan 2025 – Present**
Leadership, branding, digital communications, content strategy

- Lead media and communications division, coordinating branding, copywriting, design, and content publication across Social Media Platform.
- Manage cross-division communication for events and campaigns, ensuring consisted messaging and timely stakeholder updates.
- Headline Events and Manage Branding strategy on PPI Australia with 23.000 Followers on Instagram.

TECHNICAL PROJECTS

Suspension Pre-Heater Flow Separation CFD Analysis

- Conducting CFD Analysis to estimate the Loss due to Low Separation Efficiency on Pre-Heater Cyclone on Suspension Pre-Heater System, estimated production energy loss of approximately Rp 922,000/hour and 2.1–2.6% combustion heat loss on cement clinker production.
- CAD Design, CFD and FEA Modelling, and Calculation of Flow Deflector design to minimize Pressure Loss, Heat Loss Separation Efficiency, Heat Loss across the Combustion System.

Rotary Kiln Refractory Brick Layer

- Conducting FEA Analysis, CAD Design, and Calculation to estimate the Temperature Distribution across the kiln to estimate the Fatigue performance of the Kiln shell caused by weight and brick elongation.
- Data Modelling and Suggesting Suitable Brick Layer and Outer Shell based on on-site daily inspection.

CFD Analysis on UNSW Total Artificial Heart

- Conducting detailed CFD analysis on Nutation Movement on TAH based on Rotary Undulation Pump by using UDF.
- Code the Physics based on Journal Reference to Determine the Maximum Shear Stress and Flow Streamline across TAH Housing. Current Design has a 19 Pa Maximum Shear Stress, aimed to be less than 10 Pa in Nov 2026.

ROV Chassis Design

- 3-D Printing, Laser Cutting, and Assembly of 10+ Components of the ROV Peripherals and Chassis Design.
- Perform Design Surfacing, and Pool Testing to optimize the ROV Dynamic Performance. This ROV will compete in SAUVC 2026 Competition in Sanya, China.

TECHNICAL SKILLS

CAD/Design: SolidWorks (CSWA - Certified), Fusion 360, CATIA

Simulation: CFD, FEM, ANSYS, ABAQUS, SolidWorks Simulation **Programming/Data:** MATLAB, Python, C+, Arduino